

Proximal Balancing

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We study causal adjustment with observed pretreatment covariates X , a high-dimensional pretreatment proxy $W = (W_1, \dots, W_J)$, and an unobserved confounder. For one prespecified block j , the proposed cross-view construction learns a possibly stochastic, discrete or continuous representation $Z_j = \phi_j(V_j, \epsilon_j)$, with output dimension no larger than that of $V_j = (X, W_{-j})$, without using the held-out block W_j , and measures treatment imbalance through the conditional distribution of (X, W_j) . We prove an exact bias decomposition and show that exact cross-view balance identifies the population average treatment effect under joint latent exchangeability, held-out-view separation, overlap, and outcome-relevant completeness. When balance is approximate, a uniform outcome-relevant stability condition yields the sensitivity certificate $|\tau_{Z_j} - \tau| \leq \Gamma_j D_j(\phi_j)$. A finite-class, finite-strata learning analysis and an honest augmented inverse probability weighted estimator give an end-to-end finite-sample bound that separates representation error, hidden-confounding sensitivity, nuisance error, and evaluation-sample variation. The causal interpretation depends on the stated cross-view structural conditions; the existence of one generic correlated proxy is not sufficient by itself.

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1. Introduction

We call our method *PROxy Blockwise Exclusion* (PROBE). PROBE withholds one prespecified block of a high-dimensional pretreatment proxy, constructs an adjustment representation from the observed covariates and the remaining proxy blocks, and uses the withheld block to evaluate residual treatment imbalance. The term *exclusion* emphasizes that the evaluated proxy block is kept out of the representation input, thereby preventing literal self-copying.

1.1. Related Works

Balancing scores and covariate balancing. The propensity score is the canonical reduction of observed pretreatment covariates: under treatment ignorability given the measured covariates, conditioning on the treatment probability balances those covariates across treatment arms and identifies causal effects (Rosenbaum and Rubin, 1983). Entropy balancing and the covariate-balancing propensity score instead choose weights or treatment models that target balance directly (Hainmueller, 2012; Imai and Ratkovic, 2014). These methods motivate our optimization target, but their usual causal guarantee still requires the measured covariates to suffice for confounding control. PROBE instead allows a latent confounder and asks what a held-out proxy block can reveal about the bias remaining after reducing the other pretreatment measurements.

Causal representation learning. Representation-learning methods compress high-dimensional covariates while penalizing imbalance between treated and control representations. For example, counterfactual regression bounds counterfactual prediction error using factual prediction error and an integral probability metric between learned treatment-arm representations (Shalit et al., 2017). PROBE shares the use of a learned low-dimensional representation and a distributional balance penalty. Its distinguishing operation is blockwise exclusion: one proxy block is held out, the representation is learned from the remaining pretreatment information, and conditional treatment balance is evaluated with the held-out block.

Negative controls and proximal causal inference. Negative-control variables can reveal residual confounding because they should not be causally affected by treatment or should not causally affect the outcome, while remaining associated with hidden common causes (Lipsitch et al., 2010). Proximal causal inference turns this diagnostic idea into point identification by assigning distinct treatment-inducing and outcome-inducing proxy roles and imposing completeness and bridge conditions (Miao et al., 2018; Tchetgen Tchetgen et al., 2024). Semiparametric proximal theory then develops influence functions and efficient estimators for the resulting identifying functionals (Cui et al., 2024). PROBE takes a different operational route: the remaining proxy blocks form the input to an adjustment representation, while the held-out block measures residual treatment imbalance. An outcome-relevant completeness condition transfers exact observed balance to the potential-outcome means needed for the population average treatment effect.

Single-proxy identification. Single Proxy Control shows that one negative-control outcome can identify the effect of treatment on the treated under a counterfactual proxy condition and suitable bridge structure (Park et al., 2024). This differs from our population average treatment effect target, our latent-confounder proxy interpretation, and our use of an observable balance discrepancy. The comparison makes clear that “one proxy” does not by itself identify a common estimand. Identification depends on what the proxy measures, which conditional independence is imposed, and whether the analysis uses a bridge or an adjustment representation checked by a

held-out proxy block.

Kernel discrepancies and sensitivity certification. Maximum mean discrepancy is an integral probability metric that equals zero only for identical distributions when its kernel is characteristic (Gretton et al., 2012). PROBE uses conditional treatment-arm discrepancies of the measured covariates and the held-out block as observable magnitudes, not kernel-test p -values. Multiplying this discrepancy by an outcome-relevant stability constant yields a bias certificate and a representation-selection guarantee. The individual ingredients have close precedents in balancing, representation learning, kernel testing, and proximal inference. We therefore state the contribution through the precise assumptions, estimand, and guarantees below rather than making an unsupported priority claim.

2. Problem Setting

We observe independent copies

$$O_i = (X_i, W_i, A_i, Y_i), \quad i = 1, \dots, n, \quad (1)$$

where $X \in \mathcal{X}$ is a possibly high-dimensional vector of observed pretreatment covariates, $\mathcal{X}$ is its value space, $A \in \{0, 1\}$ is treatment, and Y is the observed outcome.

For each $a \in \{0, 1\}$, the *potential outcome* $Y(a)$ is the outcome that would be observed if treatment were set to a . Our target is the population average treatment effect

$$\tau \triangleq \mathbb{E}\{Y(1) - Y(0)\}. \quad (2)$$

Let U denote an unobserved pretreatment variable that may affect both treatment A and outcome Y . When U affects both A and Y , the observed-data distribution alone does not identify τ without additional structure. We therefore observe a proxy measurement W that contains information about U . The proxy is divided into prespecified blocks,

$$W = (W_1, \dots, W_J). \quad (3)$$

Each block may be scalar or vector-valued. A block can represent one repeated measurement, sensor channel, laboratory panel, or prespecified region of a larger measurement.

For each block index $j \in \{1, \dots, J\}$, we call W_j the *held-out proxy block* and let $\mathcal{W}_j$ denote its value space. We call the combination of all other blocks the *remaining proxy block* and write

$$W_{-j} \triangleq (W_1, \dots, W_{j-1}, W_{j+1}, \dots, W_J) \quad (4)$$

for the resulting vector of remaining proxy blocks. Let $\mathcal{V}_j$ denote the value space of the pair (X, W_{-j}) . Finally, define

$$V_j \triangleq (X, W_{-j}) \in \mathcal{V}_j. \quad (5)$$

The variable V_j is the combined vector of the observed covariates X and the remaining proxy blocks W_{-j} .

3. Identification

Let d_{V_j} denote the coordinate dimension of V_j . Fix a user-specified output dimension $d_{Z_j} \leq d_{V_j}$. Let $\epsilon_j \sim \text{Unif}(0, 1)$ be independent of $(U, V_j, W_j, A, Y(0), Y(1))$. For a measurable representation function, define

$$\phi_j : \mathcal{V}_j \times [0, 1] \longrightarrow \mathbb{R}^{d_{Z_j}}, \quad Z_j \triangleq \phi_j(V_j, \epsilon_j). \quad (6)$$

The deterministic case is included when ϕ_j ignores its second input. Let Φ_j denote a prespecified class of such representation functions.

This is the defining PROBE construction: the held-out block W_j is excluded from the input to ϕ_j and reserved for the balance condition introduced below.

3.1. Identification for a Prespecified Held-Out Block

3.1.1. Assumptions

Assumption 1 (Consistency and integrability). *For each $a \in \{0, 1\}$,*

$$Y = Y(A), \quad \mathbb{E}|Y(a)| < \infty. \quad (7)$$

Consistency connects the observed outcome Y to the potential outcome corresponding to the treatment actually received. Integrability ensures that the causal means used below are finite.

Assumption 2 (Latent exchangeability and a common held-out channel). *For each $a \in \{0, 1\}$,*

$$Y(a) \perp\!\!\!\perp A \mid (U, V_j), \quad (8)$$

$$W_j \perp\!\!\!\perp A \mid (U, V_j). \quad (9)$$

Figure 1 shows three PROBE-compatible examples and one representative standard proximal structure. In panel (d), Z^{tr} denotes a conventional treatment-inducing proxy.

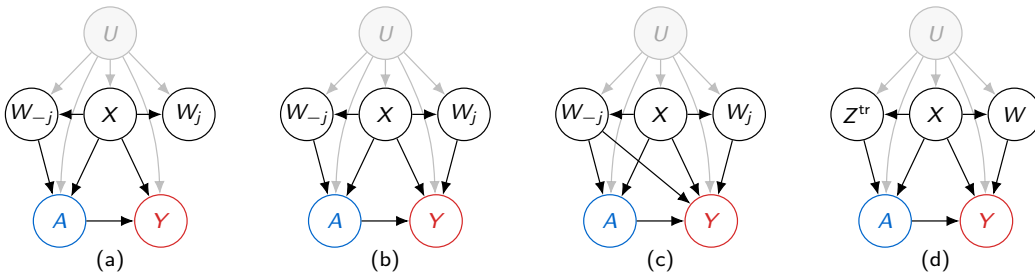


Figure 1: Panels (a)–(c) show causal structures compatible with Assumption 2. Panel (d) shows a representative standard two-proxy proximal structure. The graphs alone do not imply representation overlap, outcome-relevant completeness, or exact balance.

Assumption 3 (Representation overlap).

$$0 < P(A = 1 \mid Z_j) < 1 \quad \text{almost surely.} \quad (10)$$

We then introduce the latent and proxy objects within each representation stratum.

Definition 1 (Latent and proxy objects within a representation stratum). *Let P_{Z_j} denote the distribution of Z_j . The conditional laws below refer to fixed versions that exist under the maintained probability model. For P_{Z_j} -almost every z , define*

$$Q_{a,j,z} \triangleq \mathcal{L}(V_j, U \mid A = a, Z_j = z), \quad a \in \{0, 1\}. \quad (11)$$

Define the held-out proxy channel by

$$K_j(\cdot \mid v, u) \triangleq \mathcal{L}(W_j \mid V_j = v, U = u). \quad (12)$$

For any probability law Q on (V_j, U) and measurable $B \subseteq \mathcal{X} \times \mathcal{W}_j$, define

$$(\mathcal{K}_j Q)(B) \triangleq \int \left[\int_{\mathcal{W}_j} 1\{(x, w_j) \in B\} K_j(dw_j \mid v, u) \right] Q(dv, du), \quad v = (x, w_{-j}).$$

Equivalently, first draw $(V_j, U) = (X, W_{-j}, U) \sim Q$, then draw $W_j \sim K_j(\cdot \mid V_j, U)$, and finally retain (X, W_j) while integrating out (W_{-j}, U) . Thus, $\mathcal{K}_j Q$ is a probability law on (X, W_j) .

For each $a \in \{0, 1\}$, define

$$\mu_{a,j,z}(v, u) \triangleq \mathbb{E}\{Y(a) \mid V_j = v, U = u, Z_j = z\}. \quad (13)$$

The admissible law class is

$$\mathcal{Q}_{j,z} \triangleq \left\{ Q : Q \text{ is a probability law on } (V_j, U), \ Q \ll \mathcal{L}(V_j, U \mid Z_j = z), \right. \\ \left. \mathbb{E}_Q |\mu_{a,j,z}(V_j, U)| < \infty \text{ for } a \in \{0, 1\} \right\}. \quad (14)$$

The relation $Q \ll \mathcal{L}(V_j, U \mid Z_j = z)$ means that Q assigns zero probability to every event having zero probability under the pooled stratum law. The condition $\mathbb{E}_Q |\mu_{a,j,z}(V_j, U)| < \infty$ means that the average absolute conditional potential-outcome mean under Q is finite. For $Q \in \mathcal{Q}_{j,z}$, write

$$\mathbb{E}_Q \{\mu_{a,j,z}(V_j, U)\} \triangleq \int \mu_{a,j,z}(v, u) Q(dv, du). \quad (15)$$

Assumption 4 (Exact outcome-relevant completeness). *For P_{Z_j} -almost every z , every $Q, Q' \in \mathcal{Q}_{j,z}$, and each $a \in \{0, 1\}$,*

$$\mathcal{K}_j Q = \mathcal{K}_j Q' \implies \mathbb{E}_Q \{\mu_{a,j,z}(V_j, U)\} = \mathbb{E}_{Q'} \{\mu_{a,j,z}(V_j, U)\}. \quad (16)$$

3.1.2. Exact Identification for a Fixed Block

Observed balance alone need not remove hidden confounding. The common held-out channel and outcome-relevant completeness connect the observable balance condition to equality of the latent potential-outcome means that matter for causal identification.

Theorem 1 (Causal effect identification using a proxy). *Fix a prespecified block $j \in \{1, \dots, J\}$. Suppose Assumptions 1–4 hold for this block and its representation. If $Z_j = \phi_j(V_j, \epsilon_j)$ is constructed such that*

$$(X, W_j) \perp\!\!\!\perp A \mid Z_j, \quad (17)$$

then

$$\tau = \mathbb{E}[\mathbb{E}(Y \mid A = 1, Z_j) - \mathbb{E}(Y \mid A = 0, Z_j)]. \quad (18)$$

In plain terms, treated and untreated units have the same conditional distribution of the observed covariates X and the held-out proxy block W_j within each value of Z_j . The proof first uses the common held-out channel to factor the two observed laws through $\mathcal{K}_j$. Exact balance equates the two channel images, outcome-relevant completeness equates the corresponding potential-outcome means, and consistency and overlap yield the displayed adjustment formula. The full proof is given in Appendix B. The theorem does not claim that a representation satisfying exact balance always exists or can always be learned from a finite sample.

3.1.3. Comparison with the Canonical Two-Proxy Proximal Model

The canonical two-proxy proximal model, illustrated in Figure 1(d), prespecifies a treatment-inducing proxy Z^{tr} and an outcome-inducing proxy W (Miao et al., 2018; Tchetgen Tchetgen et al., 2024). PROBE instead places the outcome-side common-channel restriction only on the held-out block W_j . Components of the remaining block W_{-j} may affect treatment A only, outcome Y only, both, or neither, as illustrated by panels (a)–(c). This is less restrictive at the level of prespecified proxy-role exclusions, particularly when W_j is a small part of a high-dimensional W . It does not imply that the complete PROBE assumption package is weaker.

The canonical route identifies the causal effect by solving an outcome bridge equation and imposing completeness on the treatment-inducing proxy operator. PROBE instead combines observable balance with outcome-relevant completeness. A natural question is whether either framework’s sufficient conditions contain those of the other. The following proposition shows that neither package implies the other, so the two frameworks cover distinct model classes.

Proposition 1 (Non-nesting of canonical proximal and PROBE sufficient conditions). *Fix a latent variable U , observed covariates X , canonical proxy roles (Z^{tr}, W) , and a PROBE orientation (V_j, W_j, Z_j) . Let $\mathbf{C}$ denote the conjunction of the canonical proxy restrictions for (Z^{tr}, W) , existence of an integrable outcome bridge, and treatment-proxy completeness. Let $\mathbf{P}_j$ denote the conjunction of latent exchangeability, the common held-out channel, representation overlap, exact balance, and exact outcome-relevant completeness for orientation j . Within*

the class of laws satisfying consistency, integrability, and strict latent overlap $0 < P(A = 1 \mid U, X) < 1$ almost surely,

$$C \not\Rightarrow P_j, \quad P_j \not\Rightarrow C.$$

This proposition establishes non-inclusion of the two fixed sufficient-condition packages. It does not say that every method or reparameterization from one framework fails under a law covered by the other. Computationally, canonical proximal estimators solve a regularized inverse bridge or conditional-moment problem. PROBE instead optimizes an observed held-out balance criterion and then estimates the causal effect by adjustment on the learned representation. Section 4 gives the learning and AIPW details.

3.2. Identification When the Valid Held-Out Block Is Unknown

The fixed-block theorem requires the analyst to prespecify a block whose structural assumptions are valid. We now consider the leave-one-block-out orientations $j \in \{1, \dots, J\}$ without assuming that the data reveal which orientation, if any, is structurally valid. For each j , fix a candidate representation $Z_j = \phi_j(V_j, \epsilon_j)$ using the construction above.

Define the population-observable balance-screen set

$$\mathcal{B} \triangleq \left\{ j \in \{1, \dots, J\} : 0 < P(A = 1 \mid Z_j) < 1 \text{ a.s.}, \quad (X, W_j) \perp\!\!\!\perp A \mid Z_j, \right. \\ \left. \mathbb{E}[|\mathbb{E}(Y \mid A = 1, Z_j)| + |\mathbb{E}(Y \mid A = 0, Z_j)|] < \infty \right\}. \quad (19)$$

For each $j \in \mathcal{B}$, define the candidate adjustment value

$$\theta_j \triangleq \mathbb{E}[\mathbb{E}(Y \mid A = 1, Z_j) - \mathbb{E}(Y \mid A = 0, Z_j)]. \quad (20)$$

Let $\mathcal{G} \subseteq \mathcal{B}$ be the set of indices for which latent exchangeability, the common held-out channel, and exact outcome-relevant completeness in Assumptions 2 and 4 hold for that orientation.

Theorem 2 (Identification with an unknown valid block). *Suppose Assumption 1 holds and the candidate representations are fixed. Then*

$$\theta_j = \tau \quad \text{for every } j \in \mathcal{G}. \quad (21)$$

Consequently:

- (a) If $\mathcal{G} \neq \emptyset$, then $\tau \in \{\theta_j : j \in \mathcal{B}\}$. If, in addition, $\theta_j = \theta^*$ for every $j \in \mathcal{B}$, then $\tau = \theta^*$.
- (b) If $\mathcal{B} \neq \emptyset$ and $|\mathcal{G}| > |\mathcal{B}|/2$, then

$$\text{median}((\theta_j)_{j \in \mathcal{B}}) = \tau, \quad (22)$$

where one indexed value is retained for every $j \in \mathcal{B}$.

The candidate set $\{\theta_j : j \in \mathcal{B}\}$ is an existence-based containment result and is not claimed to be

sharp. Membership in $\mathcal{B}$ is determined by the population observed-data law, but exact population equality is not a finite-sample certificate. Membership in $\mathcal{G}$, its nonemptiness, and the majority-valid condition are structural properties involving the unobserved variable U and are not determined by the balance screen.

3.3. Approximate PROBE Balance and Bias Bound

Let κ_j be a bounded, measurable, characteristic kernel on (X, W_j) whose reproducing kernel Hilbert space $\mathcal{H}_j$ is separable. For probability laws P, P' on (X, W_j) , write

$$\text{MMD}_{\kappa_j}(P, P') \triangleq \|m_j(P) - m_j(P')\|_{\mathcal{H}_j}, \quad (23)$$

where $m_j(P) = \int \kappa_j(r, \cdot) P(dr)$.

Definition 2 (Held-out proxy discrepancy). *For P_{Z_j} -almost every z , define*

$$\Delta_j(z; \phi_j) \triangleq \text{MMD}_{\kappa_j} \{ \mathcal{L}(X, W_j \mid A = 1, Z_j = z), \mathcal{L}(X, W_j \mid A = 0, Z_j = z) \}, \quad (24)$$

and define

$$D_j(\phi_j) \triangleq \mathbb{E} \{ \Delta_j(Z_j; \phi_j) \}. \quad (25)$$

Proposition 2 (Zero discrepancy and exact balance). *Under representation overlap,*

$$D_j(\phi_j) = 0 \iff (X, W_j) \perp\!\!\!\perp A \mid Z_j. \quad (26)$$

The factorization used in the proof of Theorem 1 gives, almost everywhere,

$$\Delta_j(z; \phi_j) = \text{MMD}_{\kappa_j}(\mathcal{K}_j Q_{1,j,z}, \mathcal{K}_j Q_{0,j,z}). \quad (27)$$

Assumption 5 (Uniform outcome-relevant stability). *There is a finite Γ_j such that, for P_{Z_j} -almost every z , every $a \in \{0, 1\}$, and all $Q, Q' \in \mathcal{Q}_{j,z}$,*

$$|\mathbb{E}_Q \{ \mu_{a,j,z} \} - \mathbb{E}_{Q'} \{ \mu_{a,j,z} \}| \leq \Gamma_j \text{MMD}_{\kappa_j}(\mathcal{K}_j Q, \mathcal{K}_j Q'). \quad (28)$$

Uniform stability converts a difference between held-out proxy distributions into a common upper bound on the corresponding difference in potential-outcome means.

Theorem 3 (Approximate PROBE bias bound). *Under Assumptions 1–3 and Assumption 5,*

$$|\mathbb{E}[\mathbb{E}(Y \mid A = 1, Z_j) - \mathbb{E}(Y \mid A = 0, Z_j)] - \tau| \leq \Gamma_j D_j(\phi_j). \quad (29)$$

Remark 1 (Exact completeness and quantitative stability). *Stability implies exact outcome-relevant completeness, but the converse need not hold.*

Remark 2 (Sensitivity interpretation). *For a scientifically specified Γ_j ,*

$$\tau \in \left[\mathbb{E}\{\mathbb{E}(Y \mid A = 1, Z_j) - \mathbb{E}(Y \mid A = 0, Z_j)\} - \Gamma_j D_j(\phi_j), \right. \\ \left. \mathbb{E}\{\mathbb{E}(Y \mid A = 1, Z_j) - \mathbb{E}(Y \mid A = 0, Z_j)\} + \Gamma_j D_j(\phi_j) \right]. \quad (30)$$

Remark 3 (Uniformity across representations). *Learning by discrepancy requires a common budget*

$$\sup_{\phi \in \Phi_j} \Gamma_j(\phi) \leq \bar{\Gamma}_j < \infty. \quad (31)$$

4. Representation Learning and AIPW Estimation

The identification results concern one fixed representation. We now select a representation from data and estimate the adjustment functional that it induces. The analysis uses one prespecified held-out block j throughout. The AIPW construction below applies to discrete or continuous Z_j . Explicit concentration constants are subsequently given for a finite-strata specialization.

4.1. Representation Learning

Partition the observed sample into three mutually disjoint subsets,

$$\mathcal{I}_D, \quad \mathcal{I}_N, \quad \mathcal{I}_E, \quad (32)$$

with respective sizes n_D , n_N , and n_E . The discrepancy-learning sample $\mathcal{I}_D$ selects a representation. The nuisance-learning sample $\mathcal{I}_N$ estimates the outcome regressions and propensity used by AIPW. The evaluation sample $\mathcal{I}_E$ calculates the final estimate. Conditional on the preceding samples, the objects evaluated on the next sample are fixed.

Augment each fold with fresh independent randomization variables $\epsilon_{j,i}^D$, $\epsilon_{j,i}^N$, and $\epsilon_{j,i}^E$, each distributed as $\text{Unif}(0, 1)$ and independent of the observed data and of all other randomization variables. For a candidate ϕ , its representation on a unit in fold $s \in \{D, N, E\}$ is $\phi(V_{j,i}, \epsilon_{j,i}^s)$. Thus the empirical criterion below is computed from the augmented discrepancy-learning sample $\{(O_i, \epsilon_{j,i}^D) : i \in \mathcal{I}_D\}$.

For a general candidate class $\mathcal{F}_j \subseteq \Phi_j$, let $\hat{D}_j(\phi)$ be an empirical criterion estimating $D_j(\phi)$. Suppose that, with probability at least $1 - \delta_D$,

$$\sup_{\phi \in \mathcal{F}_j} |\hat{D}_j(\phi) - D_j(\phi)| \leq r_{j,n_D}. \quad (33)$$

If $\hat{\phi}_j$ is an $\varepsilon_{\text{opt},j}$ -approximate minimizer of $\hat{D}_j$, the standard empirical-risk argument gives

$$D_j(\hat{\phi}_j) \leq \inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D} + \varepsilon_{\text{opt},j}. \quad (34)$$

This oracle inequality is agnostic to whether the learned representation is discrete or continuous. Its operational content depends on constructing an empirical conditional-dependence criterion for which equation (33) can be proved. For a fresh $\epsilon_j \sim \text{Unif}(0, 1)$ independent of the causal variables, define

$$\hat{Z}_j \triangleq \hat{\phi}_j(V_j, \epsilon_j), \quad \tau_{\hat{Z}_j} \triangleq \mathbb{E}[\mathbb{E}(Y \mid A = 1, \hat{Z}_j) - \mathbb{E}(Y \mid A = 0, \hat{Z}_j)]. \quad (35)$$

This definition is used when the two conditional means are absolutely integrable under the law of $\hat{Z}_j$. If every candidate obeys the assumptions of Theorem 3 with common stability budget $\bar{\Gamma}_j$, then the same event also gives

$$|\tau_{\hat{Z}_j} - \tau| \leq \bar{\Gamma}_j \left\{ \inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D} + \varepsilon_{\text{opt},j} \right\}. \quad (36)$$

Finite-strata specialization. We now give one explicit realization of the preceding argument. Restrict to deterministic maps, identifying each such map with $v \mapsto \phi(v, 0)$ and writing it as $\phi(v)$. Assume every candidate takes values in a common finite set $\mathcal{S}_j \subset \mathbb{R}^{d_{Z_j}}$, and define $M_j \triangleq |\mathcal{S}_j|$.

Fix a finite candidate library

$$\mathcal{F}_j \subseteq \Phi_j, \quad N_j \triangleq |\mathcal{F}_j| < \infty. \quad (37)$$

For $\phi \in \mathcal{F}_j$, define

$$Z_j^\phi \triangleq \phi(X, W_{-j}) \quad (38)$$

and let $\mathcal{R}_j(\phi)$ be its set of positive-mass strata. For $z \in \mathcal{R}_j(\phi)$, write $p_{j,z}(\phi) \triangleq P(Z_j^\phi = z)$. Assume there are constants $p_{\min} > 0$ and $\eta \in (0, 1/2]$ such that every candidate and every $z \in \mathcal{R}_j(\phi)$ satisfy

$$P(Z_j^\phi = z) \geq p_{\min}, \quad (39)$$

$$\eta \leq P(A = 1 \mid Z_j^\phi = z) \leq 1 - \eta. \quad (40)$$

Assume also that the bounded kernel satisfies

$$\sup_v \sqrt{\kappa_j(v, v)} \leq \rho_j < \infty. \quad (41)$$

The two primitive held-out-block clauses of Assumption 2 do not depend on the selected candidate. Representation overlap and outcome-relevant stability must hold for every candidate-specific representation. Their stability budgets are required to obey the common bound in equation (31).

Definition 3 (Empirical held-out proxy discrepancy). *For observation $i \in \mathcal{I}_D$ and candidate*

$\phi \in \mathcal{F}_j$, write

$$Z_{j,i}^\phi \triangleq \phi(X_i, W_{-j,i}). \quad (42)$$

For treatment arm a and stratum z , define

$$N_{a,z}^\phi \triangleq \sum_{i \in \mathcal{I}_D} 1\{A_i = a, Z_{j,i}^\phi = z\}, \quad (43)$$

$$N_z^\phi \triangleq N_{0,z}^\phi + N_{1,z}^\phi, \quad (44)$$

$$\hat{p}_{j,z}(\phi) \triangleq \frac{N_z^\phi}{n_D}. \quad (45)$$

When $N_{a,z}^\phi > 0$, define the empirical kernel mean

$$\hat{m}_{a,j,z}(\phi) \triangleq \frac{1}{N_{a,z}^\phi} \sum_{i \in \mathcal{I}_D} 1\{A_i = a, Z_{j,i}^\phi = z\} \kappa_j\{(X_i, W_{j,i}), \cdot\}. \quad (46)$$

When both treatment-arm cells are nonempty, define

$$\hat{\Delta}_{j,z}(\phi) \triangleq \|\hat{m}_{1,j,z}(\phi) - \hat{m}_{0,j,z}(\phi)\|_{\mathcal{H}_j} \quad (47)$$

and

$$\hat{D}_j(\phi) \triangleq \sum_{z: N_z^\phi > 0} \hat{p}_{j,z}(\phi) \hat{\Delta}_{j,z}(\phi). \quad (48)$$

If an occupied empirical stratum contains only one treatment arm, set $\hat{D}_j(\phi) = +\infty$.

Proposition 3 (Uniform deviation for the finite-strata specialization). *Let $\delta_D \in (0, 1)$ and suppose*

$$n_D \geq \frac{8}{p_{\min} \eta} \log \left(\frac{8M_j N_j}{\delta_D} \right). \quad (49)$$

Define

$$e_{j,n_D}(\delta_D) \triangleq \rho_j \left[2 + \sqrt{2 \log \left(\frac{4M_j N_j}{\delta_D} \right)} \right] \sqrt{\frac{2}{n_D p_{\min} \eta}} \quad (50)$$

and

$$\begin{aligned} r_{j,n_D}(\delta_D) &\triangleq 2e_{j,n_D}(\delta_D) \\ &\quad + 2\rho_j \sqrt{\frac{2 \left[M_j \log 2 + \log \left(\frac{4N_j}{\delta_D} \right) \right]}{n_D}}. \end{aligned} \quad (51)$$

Under equations (39) through (41), with probability at least $1 - \delta_D$,

$$\sup_{\phi \in \mathcal{F}_j} |\widehat{D}_j(\phi) - D_j(\phi)| \leq r_{j,n_D}(\delta_D). \quad (52)$$

Definition 4 (Approximate empirical minimizer). Let $\varepsilon_{\text{opt},j} \geq 0$. The representation learner returns $\widehat{\phi}_j \in \mathcal{F}_j$ satisfying

$$\widehat{D}_j(\widehat{\phi}_j) \leq \inf_{\phi \in \mathcal{F}_j} \widehat{D}_j(\phi) + \varepsilon_{\text{opt},j}. \quad (53)$$

In this deterministic specialization, the learned representation is

$$\widehat{Z}_j \triangleq \widehat{\phi}_j(V_j). \quad (54)$$

The tolerance records optimization error and is zero for an exact empirical minimizer.

Theorem 4 (Finite-strata representation-learning oracle inequality). Suppose the conditions of Proposition 3 hold. Suppose Assumptions 1 through 2 hold for block j , and suppose representation overlap and uniform outcome-relevant stability hold for every candidate with the common budget $\bar{\Gamma}_j$. Then, with probability at least $1 - \delta_D$,

$$D_j(\widehat{\phi}_j) \leq \inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D}(\delta_D) + \varepsilon_{\text{opt},j}. \quad (55)$$

On the same event,

$$|\tau_{\widehat{Z}_j} - \tau| \leq \bar{\Gamma}_j \left[\inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D}(\delta_D) + \varepsilon_{\text{opt},j} \right]. \quad (56)$$

Remark 4 (Practical learner outside the finite-library theorem). The theorem covers hard, finite-valued representations in a finite prespecified library. A neural learner using random features, soft assignments, and Adam requires additional bounds for neural-class complexity, feature approximation, soft-to-hard transfer, and optimization error. These guarantees are not supplied by Theorem 4.

4.2. AIPW Estimation

Conditional on $\mathcal{I}_D$, the map $\widehat{\phi}_j$ is fixed. A fresh independent $\epsilon_j \sim \text{Unif}(0,1)$ produces $\widehat{Z}_j = \widehat{\phi}_j(V_j, \epsilon_j)$. Define the conditional outcome means and representation propensity

$$m_{a,j}(z) \triangleq \mathbb{E}(Y \mid A = a, \widehat{Z}_j = z), \quad (57)$$

$$e_j(z) \triangleq P(A = 1 \mid \widehat{Z}_j = z). \quad (58)$$

For the observed adjustment functional $\tau_{\hat{Z}_j}$, the following is the AIPW score:

$$\begin{aligned} \psi_j(O, \epsilon_j) &\triangleq m_{1,j}(\hat{Z}_j) - m_{0,j}(\hat{Z}_j) \\ &+ \frac{A\{Y - m_{1,j}(\hat{Z}_j)\}}{e_j(\hat{Z}_j)} - \frac{(1-A)\{Y - m_{0,j}(\hat{Z}_j)\}}{1 - e_j(\hat{Z}_j)} - \tau_{\hat{Z}_j}. \end{aligned} \quad (59)$$

When this score is square-integrable, it is the efficient influence function in the nonparametric model for $(\hat{Z}_j, A, Y)$. Bounded outcomes together with uniform representation overlap provide one sufficient condition for square integrability. Use the augmented nuisance-learning sample, with $\hat{Z}_{j,i} = \hat{\phi}_j(V_{j,i}, \epsilon_{j,i}^N)$, to estimate the nuisance functions. Denote the estimates by $\hat{m}_{0,j}$, $\hat{m}_{1,j}$, and $\hat{e}_j$.

Definition 5 (Honest AIPW estimator). *For $i \in \mathcal{I}_E$, define $\hat{Z}_{j,i} = \hat{\phi}_j(V_{j,i}, \epsilon_{j,i}^E)$ and*

$$\begin{aligned} S_j(O_i, \epsilon_{j,i}^E) &\triangleq \hat{m}_{1,j}(\hat{Z}_{j,i}) - \hat{m}_{0,j}(\hat{Z}_{j,i}) \\ &+ \frac{A_i\{Y_i - \hat{m}_{1,j}(\hat{Z}_{j,i})\}}{\hat{e}_j(\hat{Z}_{j,i})} \\ &- \frac{(1 - A_i)\{Y_i - \hat{m}_{0,j}(\hat{Z}_{j,i})\}}{1 - \hat{e}_j(\hat{Z}_{j,i})}. \end{aligned} \quad (60)$$

The honest AIPW estimator is

$$\boxed{\hat{\tau}_j^{\text{AIPW}} \triangleq \frac{1}{n_E} \sum_{i \in \mathcal{I}_E} S_j(O_i, \epsilon_{j,i}^E).} \quad (61)$$

Proposition 4 (Exact signed nuisance remainder). *Conditional on $\mathcal{I}_D$ and $\mathcal{I}_N$,*

$$\begin{aligned} &\mathbb{E}\{S_j(O, \epsilon_j)\} - \tau_{\hat{Z}_j} \\ &= \mathbb{E} \left[\{\hat{e}_j(\hat{Z}_j) - e_j(\hat{Z}_j)\} \left\{ \frac{\hat{m}_{1,j}(\hat{Z}_j) - m_{1,j}(\hat{Z}_j)}{\hat{e}_j(\hat{Z}_j)} \right. \right. \\ &\quad \left. \left. + \frac{\hat{m}_{0,j}(\hat{Z}_j) - m_{0,j}(\hat{Z}_j)}{1 - \hat{e}_j(\hat{Z}_j)} \right\} \right]. \end{aligned} \quad (62)$$

The remainder is zero if the propensity estimator is correct. It is also zero if both conditional outcome estimators are correct.

Theorem 5 (Conditional finite-sample AIPW error). *Suppose $|Y| \leq B$ almost surely. Clip the nuisance estimates so that*

$$-B \leq \hat{m}_{a,j}(z) \leq B, \quad \eta \leq \hat{e}_j(z) \leq 1 - \eta. \quad (63)$$

For a function g of $\widehat{Z}_j$, define

$$\|g\|_{2,\widehat{Z}_j} \triangleq \left[\mathbb{E}\{g(\widehat{Z}_j)^2 \mid \mathcal{I}_D, \mathcal{I}_N\} \right]^{1/2}. \quad (64)$$

After the two learning samples are observed, define

$$q_{e,j} \triangleq \|\widehat{e}_j - e_j\|_{2,\widehat{Z}_j}, \quad (65)$$

$$q_{a,j} \triangleq \|\widehat{m}_{a,j} - m_{a,j}\|_{2,\widehat{Z}_j}, \quad a \in \{0, 1\}. \quad (66)$$

For $\delta_E \in (0, 1)$, define

$$\begin{aligned} \varepsilon_{\text{AIPW},j} \triangleq & 2B \left(1 + \frac{1}{\eta} \right) \sqrt{\frac{2 \log(2/\delta_E)}{n_E}} \\ & + \frac{q_{e,j}(q_{0,j} + q_{1,j})}{\eta}. \end{aligned} \quad (67)$$

Conditional on $\mathcal{I}_D$ and $\mathcal{I}_N$, with probability at least $1 - \delta_E$ over $\mathcal{I}_E$,

$$\boxed{|\widehat{\tau}_j^{\text{AIPW}} - \tau_{\widehat{Z}_j}| \leq \varepsilon_{\text{AIPW},j}.} \quad (68)$$

4.3. End-to-End Finite-Sample Bound

Theorem 6 (Finite-strata end-to-end sensitivity bound). *Suppose the conditions of Proposition 3, Theorem 4, and Theorem 5 hold. Then, jointly over the three independent samples, with probability at least $1 - \delta_D - \delta_E$,*

$$\boxed{|\widehat{\tau}_j^{\text{AIPW}} - \tau| \leq \varepsilon_{\text{AIPW},j} + \bar{\Gamma}_j \left[\widehat{D}_j(\widehat{\phi}_j) + r_{j,n_D}(\delta_D) \right]}. \quad (69)$$

It also satisfies the oracle form

$$\boxed{\begin{aligned} |\widehat{\tau}_j^{\text{AIPW}} - \tau| \leq & \varepsilon_{\text{AIPW},j} \\ & + \bar{\Gamma}_j \left[\inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D}(\delta_D) + \varepsilon_{\text{opt},j} \right]. \end{aligned}} \quad (70)$$

Corollary 1 (Consistency). *Consider a sequence of candidate libraries, sample splits, nuisance estimators, optimization tolerances, and stability budgets satisfying Theorem 6. If*

$$\delta_{D,n} + \delta_{E,n} \longrightarrow 0, \quad (71)$$

$$\frac{\log(2/\delta_{E,n})}{n_E} \longrightarrow 0, \quad (72)$$

$$q_{e,j,n}(q_{0,j,n} + q_{1,j,n}) \xrightarrow{p} 0, \quad (73)$$

and

$$\bar{\Gamma}_{j,n} \left[\inf_{\phi \in \mathcal{F}_{j,n}} D_j(\phi) + 2r_{j,n_D}(\delta_{D,n}) + \varepsilon_{\text{opt},j,n} \right] \xrightarrow{p} 0, \quad (74)$$

then

$$\hat{\tau}_{j,n}^{\text{AIPW}} \xrightarrow{p} \tau. \quad (75)$$

Remark 5 (Interpretation and observability). *Theorem 6 separates the final error into an AIPW estimation term and a hidden-confounding sensitivity term. The stability budget $\bar{\Gamma}_j$ and the nuisance errors $q_{e,j}$, $q_{0,j}$, and $q_{1,j}$ are not generally observed. The result becomes an operational numerical certificate after these quantities are bounded by additional theory, external information, or a declared sensitivity analysis.*

Remark 6 (Scope). *The theorem uses one prespecified held-out block, hard finite strata, a finite candidate library, and an honest three-way split. It does not establish a neural-class rate, a soft-to-hard approximation bound, an Adam optimization guarantee, or a cross-fitting theorem. Experimental aggregation across several held-out orientations is also outside this fixed- j result.*

5. Experiments

5.1. Questions, Comparators, and Reporting Rules

The experiments answer three separate questions. First, does proxy-supervised representation adjustment reduce ATE error relative to adjustment using X alone? Second, do the exact-assumption designs recover the known balancing structure? Third, does the method show a useful descriptive signal in an observational benchmark with a noisy randomized reference? We report mean absolute error, root mean squared error, paired wins, held-out discrepancy, overlap diagnostics, and Monte Carlo uncertainty. Oracle procedures that observe U or the true balancing score are simulation references, not implementable competitors. The theory in Sections 3 and 4 applies to one prespecified held-out block at a time. Experimental aggregation across several orientations is reported as an empirical extension and is not presented as a separate identification theorem.

5.2. Honest Approximate SCMs

The primary approximate study fixes the causal order $U \rightarrow (X, W)$, $(X, U) \rightarrow A$, and $(A, X, U) \rightarrow Y$. Both SCMs use $\dim(X) = 20$ and $\dim(W) = 10$, with twenty independently generated data sets and nested prefixes $n \in \{100, 200, 500, 1000, 2000, 5000, 10000\}$. SCM1 uses linear proxy measurements and a nonlinear outcome component; SCM2 changes the structural equations to nonlinear transformations while preserving the same causal order and an informative monotone proxy channel. The proposed cross-mask method learns a representation from X and all but one proxy coordinate,

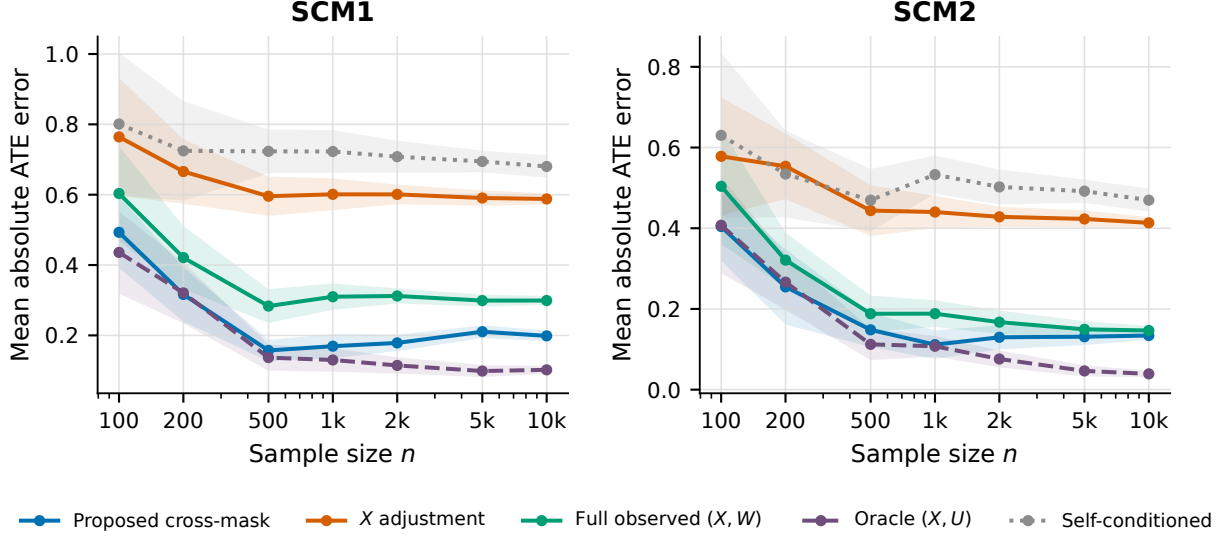


Figure 2: Honest approximate SCMs. Curves show mean absolute ATE error across twenty independently generated data sets; bands are approximate 95% Monte Carlo intervals for the mean. Prefixes are nested within a data set, so uncertainty is computed separately at each n and is not pooled across sample sizes. The proposed cross-mask learner uses one proxy coordinate for held-out auditing and averages across masks.

audits balance with the held-out coordinate, and averages over masks.

At $n = 1000$, the mean absolute errors of proposed/ X -adjustment/full- (X, W) /oracle- (X, U) were 0.169/0.601/0.310/0.130 in SCM1 and 0.112/0.440/0.188/0.107 in SCM2. At $n = 10000$, the corresponding proposed errors were 0.199 and 0.134, versus 0.588 and 0.413 for X adjustment. These are finite-sample simulation results, not a proof of the audit-view identification theorem. The self-conditioned heuristic remained substantially biased, which is consistent with the self-conditioning warning in Section 3.

5.3. Exact-Assumption Tracks A and B

Track A uses a finite balancing score, $\dim(X) = 20$, nine representation-side proxy coordinates, and one independent audit coordinate. The DGP enforces strict population overlap, exact decoding of the discrete confounding state from the representation-side view, and an injective Gaussian audit channel. At $n = 10000$ over twenty replications, mean absolute error was 0.175 for the learned audit representation, 0.433 for X adjustment, 0.186 for direct full-observed adjustment, and 0.033 for the oracle balancing score. The primary comparison with X passed, while the prespecified secondary superiority criterion against full-observed adjustment failed.

Track B uses a continuous exact balancing score and a linear representation class matched to that score. At $n = 10000$, mean absolute error was 0.0748 for the learned representation, 0.1583 for fitted X adjustment, and 0.1342 for fitted full-observed adjustment. The learned method beat X in 20/20 and full-observed adjustment in 19/20 paired replications. Its mean rank-recovery Spearman correlation was 0.915. The held-out conditional-dependence diagnostic used finite random features

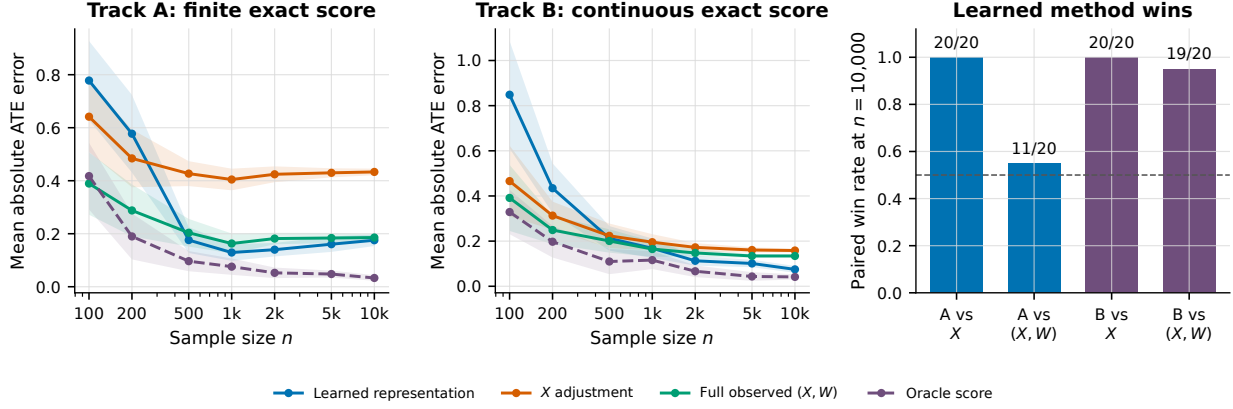


Figure 3: Exact-assumption experiments. The first two panels report mean absolute ATE error with approximate 95% Monte Carlo intervals over twenty replications. The third panel reports paired win rates at $n = 10,000$, where a win means that the learned representation has smaller absolute error than the indicated comparator. Track A satisfies the primary comparison with X but not the prespecified secondary superiority criterion against full observed adjustment. Track B records rank-recovery Spearman correlation 0.915 and held-out finite-RFF conditional-dependence surrogate 0.00691.

and is a surrogate, not an exact population KCI test; the design also covers one prespecified scalar- U SCM family.

5.4. WSC Observational Benchmark

The WSC study forms an observational subset of 1,105 participants whose randomized mathematics assignment agrees with pretreatment preference, and targets the population ATE in the original 2,200 randomized participants. The full randomized difference in means is 0.760 with standard error 0.151 and serves only as a noisy benchmark. Across ten algorithmic split repetitions on this one fixed data set, the unconstrained cross-mask estimate averaged 1.302 with split standard deviation 0.065 and absolute benchmark gap 0.541, compared with gaps 2.690 for the naive observational contrast, 1.704 for direct X adjustment, and 0.672 for direct (X, W) adjustment. The unconstrained cross-mask method is the proposed WSC method shown in Figure 4. Its practical-overlap gate passed 0/10 repetitions. The benchmark therefore supplies descriptive evidence only, not a validated causal-performance claim, causal ground truth, or a test of proxy validity.

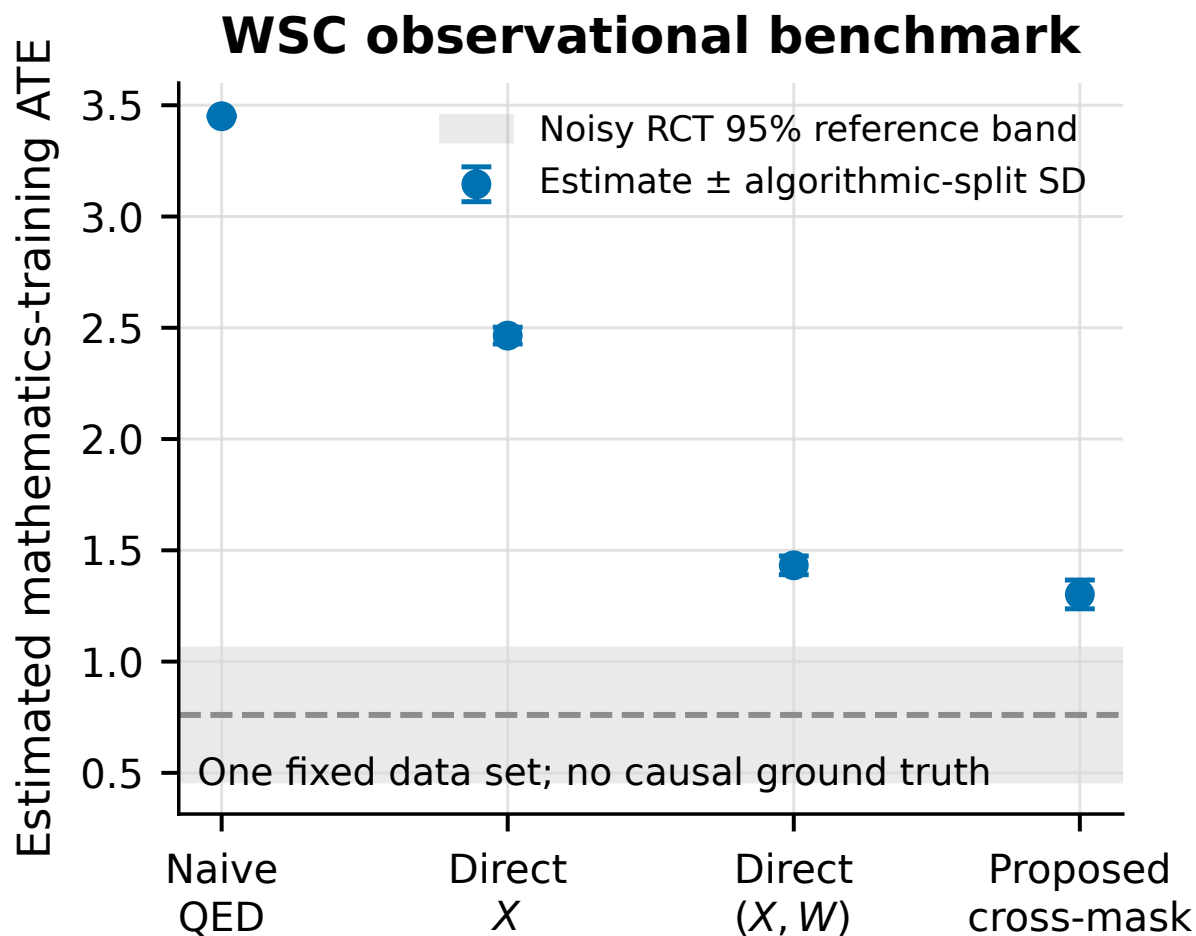


Figure 4: WSC observational benchmark. Estimates are shown relative to the noisy randomized reference. Error bars for fitted methods are standard deviations across ten algorithmic splits of one fixed data set, not Monte Carlo uncertainty. The benchmark has no causal ground truth and reports only the unconstrained cross-mask learner as the proposed method.

6. Discussion and Limitations

6.1. What the Method Delivers

For one prespecified block j , the canonical method uses W_{-j} to construct an outcome-free representation and uses the held-out block W_j to measure the residual imbalance of (X, W_j) across treatment arms within that representation. Under uniform outcome-relevant stability, the product $\Gamma_j D_j(\phi_j)$ is a sensitivity certificate for hidden-confounding bias. This is stronger than reporting a conditional-independence test alone because it targets a causal error scale, but weaker than point identification when $D_j(\phi_j) > 0$ because Γ_j is not generally identified from the observed distribution.

6.2. Testable and Structural Components

Held-out joint balance, empirical overlap, and minimum stratum mass are observable properties. Joint latent exchangeability, separation of the held-out proxy block, outcome-relevant completeness, and outcome-relevant stability are structural assumptions about the full-data law. Exact balance is therefore an observable population condition that can only be assessed up to sampling error, while its causal interpretation still requires structural assumptions. A large held-out discrepancy can falsify exact balance; a small test statistic cannot prove exact balance, completeness, or stability.

6.3. Attainability and Realized Bias

The exact identification theorem is conditional on the existence of a representation $Z_j = \phi_j(V_j, \epsilon_j)$ that attains exact held-out proxy balance. The theorem does not guarantee such attainability in a restricted representation class. In the approximate regime, reducing $D_j(\phi_j)$ reduces the certified worst-case bias under a common stability budget, but it need not reduce the realized bias when candidate representations have different stability constants or when the structural stability condition fails.

6.4. Learning and Inference Gaps

The current explicit finite-sample theorem covers finite candidate libraries with hard finite strata and fixed-fold evaluation. The neural experiments use soft assignments and covariance-weighted surrogate objectives that are not algebraically identical to the canonical MMD aggregation. A neural-class generalization bound, a soft-to-hard guarantee, and valid joint inference after adaptive representation selection remain open. These gaps prevent the manuscript from describing every neural result as a direct verification of the theory, even when its empirical error is lower.

6.5. Audit Views and Practical Scope

PROBE uses one prespecified proxy block for held-out evaluation and the remaining high-dimensional measurement for representation learning. This design blocks literal self-copying of the evaluated block. It does not by itself establish the required held-out-block structure, so point

identification still needs the structural assumptions in Section 3. The experiments repeat the procedure across several held-out orientations and aggregate their estimates as an empirical extension. The population median result in Theorem 2 justifies aggregation only when a strict majority of the balance-screened orientations are structurally valid; the current finite-sample learning and inference theory remains fixed- j . The WSC experiment further shows that an encouraging benchmark gap can coexist with a failed practical-overlap rule. Real-data evidence must therefore report overlap, held-out discrepancies, and benchmark uncertainty without treating any one diagnostic as ground truth.

6.6. Scope of the Contribution

The defensible contribution is the combination of a blockwise proxy-supervised representation, an observable conditional discrepancy, an exact identification theorem, a stability-indexed approximate guarantee, and an honest finite-sample estimation analysis. The paper does not claim that one generic correlated proxy point-identifies the ATE, that every smaller discrepancy has smaller realized bias without a common stability budget, or that the current neural optimizer reaches the population optimum. A systematic literature audit and broader external validation are required before any firstness or universality claim.

7. Conclusion

Single proxy balancing asks what a structured high-dimensional pretreatment proxy can reveal about hidden-confounding bias after learning a low-dimensional adjustment representation. For one prespecified held-out block j , PROBE learns $Z_j = \phi_j(V_j, \epsilon_j)$ and evaluates treatment balance through (X, W_j) . Exact held-out proxy balance identifies the ATE under the stated structural assumptions, while approximate balance yields the sensitivity certificate

$$|\mathbb{E}\{\mathbb{E}(Y \mid A = 1, Z_j) - \mathbb{E}(Y \mid A = 0, Z_j)\} - \tau| \leq \Gamma_j D_j(\phi_j).$$

Empirical discrepancy minimization and honest AIPW estimation give an end-to-end bound that separates learning, hidden-confounding, nuisance, and evaluation errors. The current simulations support the method in exact and approximate designs while exposing limitations from unattainable balance, weak proxy channels, surrogate mismatch, and practical-overlap loss. Neural-class generalization, soft-to-hard transfer, stability calibration, and external validation remain necessary for a submission-ready paper.

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A. Simulation Details

The existing project-local experiments will be documented by recording each data-generating process, parameter value, sample size, replication count, and random seed policy. Every manuscript figure and table will be linked to the script and machine-readable result from which it was produced.

The simulation audit will separate observed quantities from oracle quantities. In particular, any result that uses U or a direction constructed from U will be labeled as an oracle diagnostic and will not be presented as the output of a learnable procedure.

B. Proofs

B.1. Proofs for Section 3

Exact identification.

Proof of Theorem 1. Because Z_j is constructed from V_j and the independent variable ϵ_j , the two clauses of Assumption 2 remain valid after additionally conditioning on Z_j . Thus, for each $a \in \{0, 1\}$,

$$\mathbb{E}\{Y(a) \mid A, V_j, U, Z_j\} = \mathbb{E}\{Y(a) \mid V_j, U, Z_j\}, \quad (76)$$

$$\mathcal{L}(W_j \mid A, V_j, U, Z_j) = K_j(\cdot \mid V_j, U). \quad (77)$$

Integrating the second identity over $Q_{b,j,z}$ and using the definition of $\mathcal{K}_j$ gives

$$\mathcal{L}(X, W_j \mid A = b, Z_j = z) = \mathcal{K}_j Q_{b,j,z}. \quad (78)$$

Exact balance in equation (17) therefore implies

$$\mathcal{K}_j Q_{1,j,z} = \mathcal{K}_j Q_{0,j,z} \quad (79)$$

for P_{Z_j} -almost every z . Representation overlap gives $Q_{b,j,z} \ll \mathcal{L}(V_j, U \mid Z_j = z)$ for $b \in \{0, 1\}$. Assumption 1, equation (76), and Jensen's inequality give the required integrability, so $Q_{b,j,z} \in \mathcal{Q}_{j,z}$ almost everywhere. Assumption 4 therefore gives, for each $a \in \{0, 1\}$,

$$\mathbb{E}_{Q_{1,j,z}}\{\mu_{a,j,z}(V_j, U)\} = \mathbb{E}_{Q_{0,j,z}}\{\mu_{a,j,z}(V_j, U)\}. \quad (80)$$

By equation (76) and iterated expectation, for each $b \in \{0, 1\}$,

$$\mathbb{E}\{Y(a) \mid A = b, Z_j = z\} = \mathbb{E}_{Q_{b,j,z}}\{\mu_{a,j,z}(V_j, U)\}. \quad (81)$$

Equation (80) shows that the conditional mean of $Y(a)$ given (A, Z_j) does not depend on A . Consistency then gives

$$\begin{aligned} \mathbb{E}(Y \mid A = a, Z_j = z) &= \mathbb{E}\{Y(a) \mid A = a, Z_j = z\} \\ &= \mathbb{E}\{Y(a) \mid Z_j = z\}. \end{aligned} \quad (82)$$

Integrating over P_{Z_j} yields equation (18). $\square$

Non-nesting of canonical proximal and PROBE sufficient conditions.

Proof of Proposition 1. We give one finite-state law for each direction. In both constructions, X is constant, $Y = Y(A)$, and every treatment draw is generated by an exogenous randomizer independent of all proxy noises and outcome noises.

First, we show $C \not\Rightarrow P_j$. Let

$$U \sim \text{Bernoulli}(1/2), \quad P(A = 1 \mid U = u) = \frac{1}{4} + \frac{1}{2}u,$$

and let

$$Z^{\text{tr}} = U \oplus E_Z, \quad W = W_j = U \oplus E_W,$$

where E_Z and E_W are mutually independent Bernoulli(1/4) variables, independent of the treatment randomizer. Set $Y(a) = a + U$ and $V_j = Z^{\text{tr}}$. Consistency, integrability, and strict latent overlap hold. Independence of the exogenous noises gives the three canonical proxy-validity conditions in C .

The function

$$h(w, a) = a + 2w - \frac{1}{2}$$

is an outcome bridge because $\mathbb{E}(W \mid U = u) = 1/4 + u/2$, so $\mathbb{E}\{h(W, a) \mid U = u\} = a + u$. Iterated expectation gives the required observed bridge equation. The binary channel from U to Z^{tr} has matrix

$$\begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}$$

and determinant 1/2. Conditional on either treatment arm, both latent states have positive probability, and the corresponding unnormalized conditional-expectation matrix has determinant 3/32. It therefore has full column rank, which proves the canonical treatment-proxy completeness condition.

Now consider any measurable randomized representation $Z_j = \phi_j(V_j, \epsilon_j)$, where ϵ_j is independent of all preceding variables. The Markov relation $U \rightarrow V_j \rightarrow Z_j$ holds. Conditional on (U, Z_j) , treatment and W_j are independent, and

$$\mathbb{E}(A \mid U, Z_j) = \mathbb{E}(W_j \mid U, Z_j) = \frac{1}{4} + \frac{1}{2}U.$$

The conditional covariance decomposition gives

$$\text{Cov}(A, W_j \mid Z_j) = \frac{1}{4} \text{Var}(U \mid Z_j).$$

Because $P(U = 1 \mid V_j = 0) = 1/4$ and $P(U = 1 \mid V_j = 1) = 3/4$,

$$P(U = 1 \mid Z_j) = \mathbb{E}\{P(U = 1 \mid V_j) \mid Z_j\} \in [1/4, 3/4]$$

almost surely. Hence $\text{Var}(U \mid Z_j) \geq 3/16$ and

$$\text{Cov}(A, W_j \mid Z_j) \geq \frac{3}{64} > 0.$$

Exact balance therefore fails for every such representation. This proves $\mathbf{C} \not\Rightarrow \mathbf{P}_j$.

Second, we show $\mathbf{P}_j \not\Rightarrow \mathbf{C}$. Let $U = (U_1, U_2)$, where U_1 and U_2 are independent Bernoulli(1/2) variables. Let Z^{tr} and W be separate observed coordinates that agree almost surely with U_1 , and set

$$V_j = Z^{\text{tr}} = U_1, \quad W_j = W = U_1, \quad Z_j = U_1.$$

Define

$$P(A = 1 \mid U_1, U_2) = \frac{3}{20} + \frac{1}{5}U_1 + \frac{2}{5}U_2,$$

and let ξ be an independent mean-zero Rademacher variable, also independent of the treatment randomizer. Set

$$Y(a) = a + U_1 + (1 + U_2)\xi.$$

The four latent treatment probabilities are 3/20, 11/20, 7/20, and 15/20, so strict latent overlap holds. Latent exchangeability follows from the independent outcome noise, and the common held-out channel holds because W_j is deterministic given U . Moreover,

$$P(A = 1 \mid Z_j = 0) = \frac{7}{20}, \quad P(A = 1 \mid Z_j = 1) = \frac{11}{20},$$

so representation overlap holds. Because $W_j = Z_j$ and X is constant, exact balance also holds.

It remains to verify the all- Q outcome-relevant completeness condition. Within a stratum $Z_j = z$, every admissible law $Q \in \mathcal{Q}_{j,z}$ is supported on $V_j = U_1 = z$, while U_2 may vary. On this support,

$$\mu_{a,j,z}(v, u) = a + z.$$

Thus, for any admissible Q, Q' and either potential treatment level,

$$\mathcal{K}_j Q = \mathcal{K}_j Q' \implies \mathbb{E}_Q\{\mu_{a,j,z}(V_j, U)\} = a + z = \mathbb{E}_{Q'}\{\mu_{a,j,z}(V_j, U)\}.$$

This proves all conditions in $\mathbf{P}_j$.

The outcome bridge $h(w, a) = a + w$ exists because

$$\mathbb{E}(Y \mid Z^{\text{tr}} = z, A = a) = a + z = \mathbb{E}\{h(W, a) \mid Z^{\text{tr}} = z, A = a\}.$$

Canonical treatment-proxy completeness nevertheless fails. In the treated arm,

$$P(U_2 = 1 \mid U_1 = 0, A = 1) = \frac{11}{14}, \quad P(U_2 = 1 \mid U_1 = 1, A = 1) = \frac{15}{22}.$$

Define the square-integrable, nonzero function

$$g_1(U) = U_2 - \begin{cases} 11/14, & U_1 = 0, \\ 15/22, & U_1 = 1. \end{cases}$$

Then

$$\mathbb{E}\{g_1(U) \mid Z^{\text{tr}}, A = 1\} = \mathbb{E}\{g_1(U) \mid U_1, A = 1\} = 0$$

almost surely, although $g_1(U)$ is not almost surely zero. Hence the canonical completeness condition fails, proving $P_j \not\Rightarrow C$. $\square$

Identification with an unknown valid block.

Proof of Theorem 2. Fix any $j \in \mathcal{G}$. By the definition of $\mathcal{G}$, the latent exchangeability, common held-out channel, and exact outcome-relevant completeness conditions hold for orientation j . Because $j \in \mathcal{B}$, representation overlap and exact held-out proxy balance also hold. Assumption 1 and Theorem 1 therefore give

$$\theta_j = \tau. \tag{83}$$

This conclusion holds for every $j \in \mathcal{G}$. If $\mathcal{G} \neq \emptyset$, at least one element of $\{\theta_j : j \in \mathcal{B}\}$ therefore equals τ . If every indexed value equals the same θ^* , then necessarily $\tau = \theta^*$. Finally, if $|\mathcal{G}| > |\mathcal{B}|/2$, strictly more than half of the indexed values in $(\theta_j)_{j \in \mathcal{B}}$ equal τ . Consequently, the middle indexed value when $|\mathcal{B}|$ is odd, and both middle indexed values when $|\mathcal{B}|$ is even, equal τ . Hence the median of the indexed list, with multiplicities retained, is τ . $\square$

Zero discrepancy and exact balance.

Proof of Proposition 2. The nonnegative random variable $\Delta_j(Z_j; \phi_j)$ has expectation zero exactly when it is zero almost surely. Characteristicness converts zero MMD into equality of the two conditional laws, and overlap makes this equality equivalent to conditional independence. $\square$

Supporting bias decomposition. For this supporting calculation, define $\pi_j(z) \triangleq P(A = 1 \mid Z_j = z)$. For each $a \in \{0, 1\}$ and P_{Z_j} -almost every z , define

$$\delta_{a,j}(z) \triangleq \mathbb{E}_{Q_{1,j,z}}\{\mu_{a,j,z}(V_j, U)\} - \mathbb{E}_{Q_{0,j,z}}\{\mu_{a,j,z}(V_j, U)\}. \tag{84}$$

Under Assumptions 1–3, consistency and equation (76) give, for P_{Z_j} -almost every z ,

$$\mathbb{E}(Y \mid A = a, Z_j = z) = \mathbb{E}_{Q_{a,j,z}}\{\mu_{a,j,z}(V_j, U)\}. \tag{85}$$

Conditional expectation over A gives

$$\begin{aligned} \mathbb{E}\{Y(a) \mid Z_j = z\} &= \pi_j(z) \mathbb{E}_{Q_{1,j,z}}\{\mu_{a,j,z}\} \\ &\quad + \{1 - \pi_j(z)\} \mathbb{E}_{Q_{0,j,z}}\{\mu_{a,j,z}\}. \end{aligned} \tag{86}$$

Subtracting the two conditional expressions and collecting terms gives, for P_{Z_j} -almost every z ,

$$\begin{aligned} & \mathbb{E}(Y \mid A = 1, Z_j = z) - \mathbb{E}(Y \mid A = 0, Z_j = z) \\ &= [\mathbb{E}\{Y(1) \mid Z_j = z\} - \mathbb{E}\{Y(0) \mid Z_j = z\}] \\ &= \{1 - \pi_j(z)\}\delta_{1,j}(z) + \pi_j(z)\delta_{0,j}(z). \end{aligned} \tag{87}$$

Approximate PROBE bias bound.

Proof of Theorem 3. Representation overlap gives $Q_{b,j,z} \ll \mathcal{L}(V_j, U \mid Z_j = z)$ for $b \in \{0, 1\}$. Assumption 1, equation (76), and Jensen's inequality give the required integrability, so $Q_{b,j,z} \in \mathcal{Q}_{j,z}$ almost everywhere. Assumption 5 and equation (27) therefore give, for each $a \in \{0, 1\}$ and almost every z ,

$$|\delta_{a,j}(z)| \leq \Gamma_j \Delta_j(z; \phi_j). \tag{88}$$

The bounded kernel and finite Γ_j imply $D_j(\phi_j) < \infty$, so the right-hand side of equation (87) is integrable. Assumption 1 makes the conditional potential-outcome contrast integrable. Equation (87) therefore also makes the observed conditional contrast integrable. We may now integrate that identity and use the triangle inequality and equation (88) to obtain

$$\begin{aligned} & |\mathbb{E}[\mathbb{E}(Y \mid A = 1, Z_j) - \mathbb{E}(Y \mid A = 0, Z_j)] - \tau| \\ & \leq \mathbb{E}[|1 - \pi_j(Z_j)|\delta_{1,j}(Z_j)| + \pi_j(Z_j)|\delta_{0,j}(Z_j)|] \end{aligned} \tag{89}$$

$$\leq \Gamma_j \mathbb{E}\{\Delta_j(Z_j; \phi_j)\}. \tag{90}$$

□

B.2. Proofs for Section 4

Uniform discrepancy deviation.

Proof of Proposition 3. Equation (49) and a multiplicative Chernoff bound imply that every candidate treatment-by-stratum cell contains at least $n_D p_{\min} \eta / 2$ observations except on an event of probability at most $\delta_D / 4$. Conditional on the cell assignments, bounded Hilbert-space mean concentration and a union bound over at most $2M_j N_j$ cells give

$$|\hat{\Delta}_{j,z}(\phi) - \Delta_j(z; \phi)| \leq 2e_{j,n_D}(\delta_D) \tag{91}$$

simultaneously, with failure probability at most $\delta_D / 2$. A multinomial L_1 concentration bound gives

$$\sum_z |\hat{p}_{j,z}(\phi) - p_{j,z}(\phi)| \leq \sqrt{\frac{2[M_j \log 2 + \log(4N_j/\delta_D)]}{n_D}} \tag{92}$$

simultaneously over candidates, with failure probability at most $\delta_D / 4$. On the intersection of the three events, use $0 \leq \Delta_j(z; \phi) \leq 2\rho_j$ and the triangle inequality to obtain equation (52). The three failure probabilities sum to δ_D . □

Representation-learning oracle inequality.

Proof of Theorem 4. On the uniform-deviation event,

$$D_j(\hat{\phi}_j) \leq \hat{D}_j(\hat{\phi}_j) + r_{j,n_D}(\delta_D) \quad (93)$$

$$\leq \inf_{\phi \in \mathcal{F}_j} \hat{D}_j(\phi) + \varepsilon_{\text{opt},j} + r_{j,n_D}(\delta_D) \quad (94)$$

$$\leq \inf_{\phi \in \mathcal{F}_j} D_j(\phi) + 2r_{j,n_D}(\delta_D) + \varepsilon_{\text{opt},j}. \quad (95)$$

This proves equation (55). Conditional on $\mathcal{I}_D$, the selected map is fixed. Theorem 3 and the common stability budget give $|\tau_{\hat{Z}_j} - \tau| \leq \bar{\Gamma}_j D_j(\hat{\phi}_j)$. Substitution proves equation (56). $\square$

Exact signed nuisance remainder.

Proof of Proposition 4. Fix $\hat{Z}_j = z$. Taking the conditional expectation of the score and using equation (58) gives

$$\begin{aligned} \mathbb{E}\{S_j(O, \epsilon_j) \mid \hat{Z}_j = z\} &= \hat{m}_{1,j}(z) - \hat{m}_{0,j}(z) \\ &\quad + \frac{e_j(z)\{m_{1,j}(z) - \hat{m}_{1,j}(z)\}}{\hat{e}_j(z)} \\ &\quad - \frac{\{1 - e_j(z)\}\{m_{0,j}(z) - \hat{m}_{0,j}(z)\}}{1 - \hat{e}_j(z)}. \end{aligned} \quad (96)$$

Subtract $m_{1,j}(z) - m_{0,j}(z)$ and collect the two nuisance errors. Averaging the resulting identity over $\hat{Z}_j$ proves equation (62). $\square$

Conditional finite-sample AIPW error.

Proof of Theorem 5. The boundedness and clipping conditions imply

$$|S_j(O, \epsilon_j)| \leq 2B \left(1 + \frac{1}{\eta}\right). \quad (97)$$

Conditional on the two learning samples, Hoeffding's inequality bounds the difference between the evaluation average and its conditional expectation by the first term in equation (67). Proposition 4, Cauchy–Schwarz, and the propensity clipping bound control the absolute conditional remainder by

$$\frac{q_{e,j}(q_{0,j} + q_{1,j})}{\eta}. \quad (98)$$

The triangle inequality proves equation (68). $\square$

End-to-end sensitivity bound.

Proof of Theorem 6. The triangle inequality gives

$$|\hat{\tau}_j^{\text{AIPW}} - \tau| \leq |\hat{\tau}_j^{\text{AIPW}} - \tau_{\hat{Z}_j}| + |\tau_{\hat{Z}_j} - \tau|. \quad (99)$$

Theorem 5 bounds the first term. Theorem 3 and uniform candidate stability bound the second term by $\bar{\Gamma}_j D_j(\hat{\phi}_j)$. Proposition 3 gives

$$D_j(\hat{\phi}_j) \leq \hat{D}_j(\hat{\phi}_j) + r_{j,n_D}(\delta_D), \quad (100)$$

which proves equation (69). Equation (55) proves the oracle form. A union bound over the discrepancy-learning and evaluation events gives the stated joint probability. $\square$

Consistency.

Proof of Corollary 1. The first three conditions make the evaluation-sample and nuisance-product components of $\varepsilon_{\text{AIPW},j}$ converge to zero. Equation (74) makes the stability-weighted representation term vanish. The failure probability of Theorem 6 also converges to zero, which proves convergence in probability. $\square$